



## Review

# Three-Dimensional human motion analysis using LiDAR technology: A systematic review

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## ABSTRACT

3D Light Detection and Ranging (LiDAR) has gained increasing attention in the field of human motion analysis due to its capability to non-invasively capture dynamic 3D pose data. However, accurate extraction of motion information from sparse and unordered point clouds remains a considerable challenge. This study aimed to systematically review 3D LiDAR data processing methods for human motion analysis. The search employed five databases (Web of Science, Scopus, Medline, PubMed and Embase) to identify methods for 3D human motion detection using LiDAR data. Following screening of 752 articles, a total of 38 studies were included. Convolutional Neural Networks (CNNs) were the most commonly used algorithm for human detection from LiDAR data, while the PointNet family was widely employed in point cloud feature extraction for human joint position estimation. The CNN-based YOLOv3 model achieved the highest human detection precision (97.7%), while a fusion approach integrating 3D LiDAR data with four inertial measurement units achieved the lowest Mean Joint Position Error (30.0 mm). Use of a CNN-based Federated Learning framework with Dynamic Layer Sharing achieved the highest activity recognition accuracy (98%). Integrating deep learning techniques enables effective extraction of spatiotemporal motion patterns from raw, sparse, and unordered point clouds, enhancing LiDAR-based human motion analysis. Significant challenges still remain in improving the usability of 3D LiDAR systems, particularly in handling multi-person scenarios and ensuring robust human motion analysis in the presence of LiDAR occlusion.

## 1. Introduction

Human motion analysis, which has played a crucial role in our understanding of human joint function, is routinely applied in healthcare, sports science, defense, gaming, film and animation industries, as well as robotics (Haque et al., 2020; Lui et al., 2025; van der Kruk and Reijne, 2018). Since measurements of joint movement can provide objective information on neuromusculoskeletal function, motion analysis continues to play a major role in musculoskeletal disease diagnostics, surgical planning, rehabilitation, ergonomics, human performance and injury risk assessment, and in evaluating human-machine interaction.

Marker-based optoelectronic motion capture is regarded as the gold standard in clinical motion analysis. However, commercially available mocap systems are costly, require significant setup time and are typically confined to specialized motion analysis laboratories (Sun et al., 2023). In contrast, wearable sensors such as inertial measurement units (IMUs) can record acceleration, gyroscopic and magnetometer data, can

be employed independent of environmental factors such as lighting, temperature, and external conditions, and are not restricted by capture volume. While IMUs are commonly used to measure relative angular motion of segments and joints, they cannot be used to delineate absolute position of body segments in space (López-Nava and Muñoz-Meléndez, 2016). Their motion measurement also relies on careful sensor placement and alignment with bony landmarks, sensor fusion algorithms, and drift correction, making precise 3D motion estimation challenging in practice (Yi et al., 2021).

Video-based motion analysis systems, which include red-green-blue (RGB) cameras from smartphones and consumer webcams, can be employed to capture full-body motion from external viewpoints without the use of physical markers. But these systems, which typically require two or more video cameras, can raise privacy concerns, particularly when employed in home or healthcare settings (Martinez-Martin et al., 2021; Uhlrich et al., 2023; Zheng et al., 2024). Furthermore, video-based systems inherently rely on visible light, making them sensitive

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to changes in illumination conditions such as low light, which may degrade tracking robustness (Tatana et al., 2025).

In recent years, three-dimensional Light Detection and Ranging (3D LiDAR) has seen increased uptake in human motion capture applications due to its ability to support contactless acquisition of full-body 3D motion and environmental information (Ul Alam et al., 2021). This technology can provide continuous spatial measurements without dependence on wearable markers or sensors, and capture absolute three-dimensional positions of the body rather than relative segment orientations. LiDAR does not record identifiable skin or facial features, and so partially mitigates the privacy concerns associated with conventional video-based technologies. Because LiDAR can record the 3D position and orientation of all objects in the sensor field of view, it provides capability for evaluating the way people move and interact with their environment.

While LiDAR provides rich three-dimensional spatial information, the point clouds captured by this technology are typically sparse, unordered, and unstructured, which poses substantial challenges in the calculation of accurate human motion information. A range of machine learning and deep learning modeling frameworks have been developed to convert 3D LiDAR data into human motion, including symmetric aggregation-based networks to address unordered inputs, graph neural networks (GNNs) to represent topological relationships, convolutional neural networks (CNNs) to extract spatial information from projected representations, and recurrent neural networks (RNNs) to incorporate temporal continuity (Cong et al., 2022; Li et al., 2022; Lin et al., 2024).

Despite increasing use of 3D LiDAR in human motion analysis applications, data processing methods for extracting human kinematics information from sparse point clouds remain under-researched. The aim of this study was to systematically review methodological approaches for human motion analysis using LiDAR technology, and their accuracy in human detection, joint position estimation, and activity recognition. The findings of this study may support LiDAR use for mobility measurement, including remote monitoring, telemedicine and rehabilitation in the home setting.

## 2. Methods

This systematic review was conducted in accordance with the PRISMA guidelines (Page et al., 2021). Studies were selected based on eligibility criteria with sequential title and abstract screening, followed by full-text review.

### 2.1. Database search strategy

A literature search was conducted to identify articles published between January 1, 2000 and February 13, 2025, using the following five electronic databases: (1) Web of Science, (2) Scopus, (3) Medline, via Ovid, (4) PubMed, and (5) Embase, via Ovid. A search strategy was used to minimize possible missing of relevant studies: (“LiDAR”) AND (“human activity recognition” OR “human motion” OR “joint angle” OR “human movement” OR “human kinematics” OR “human biomechanics” OR “human joint motion” OR “human joint movement”).

### 2.2. Selection criteria

After the removal of duplicates, all titles and abstracts were screened to ensure they involved human motion analysis using LiDAR data. Exclusion criteria included (i) non-English studies, (ii) thesis or review articles, (iii) studies that did not involve living human participants, and (iv) studies that did not report any quantitative metrics used to evaluate the accuracy or reliability of their approach (Fig. 1).

### 2.3. Quality assessment

The quality of all included studies was evaluated using a customized quality assessment based on the Downs & Black, QUADAS-2 and PROBAST checklists (Downs and Black, 1998; Moons et al., 2019; Whiting et al., 2011) (Table 1). The quality assessment questions addressed key characteristics of the included studies, including clarity of aims, transparency of data sources, reliability of annotations, reproducibility of methodological design, error analysis, validation strategies, and model

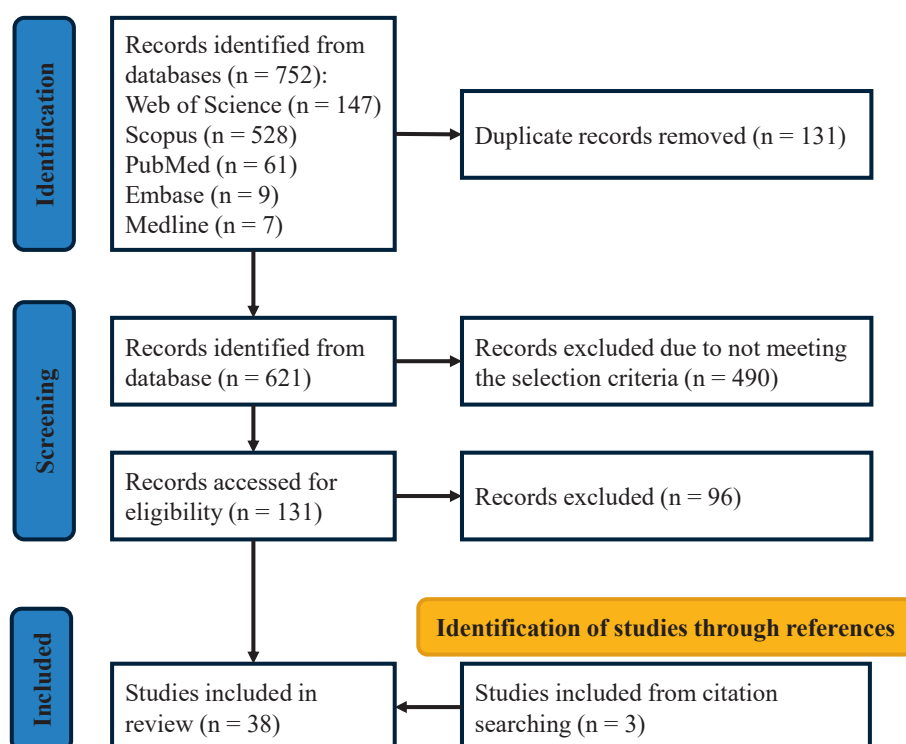


Fig. 1. PRISMA flow diagram for systematic review.

**Table 1**  
Quality assessment checklist.

| No. | Questions                                                                                         |
|-----|---------------------------------------------------------------------------------------------------|
| 1   | Was the study's aim clearly defined?                                                              |
| 2   | Were the datasets or data collection procedures clearly described and justified?                  |
| 3   | Were ground-truth annotations obtained using reliable reference systems?                          |
| 4   | Was the experimental design appropriate to answer the stated research?                            |
| 5   | Was the methodology described in sufficient detail to ensure clarity?                             |
| 6   | Were evaluation metrics clearly defined?                                                          |
| 7   | Was internal validation or external validation on independent datasets performed?                 |
| 8   | Did the study compare the proposed method with existing baselines or state-of-the-art approaches? |
| 9   | Were code, models, or parameters provided to support reproducibility?                             |

comparisons. There were 9 questions in total, and each was scored 0, 1, or 2 which corresponded to not addressed, partially addressed, or fully addressed, respectively. Quality scores were collated, and their mean and range were calculated. High methodological quality was defined as a score of  $\geq 16$  (to a maximum of 18), moderate quality was defined as a score of  $\geq 12$  and  $< 16$ , and low quality was defined as a score of  $< 12$ . Two reviewers participated in the quality assessment independently, and any disagreement in the scores was resolved by discussion.

**3. Results**

**3.1. Search outcome and quality assessment**

A total of 752 studies were identified from the initial search, of which 131 duplicates were removed. After title and abstract screening, 131 eligible studies were retrieved for the full-text screening based on the selection criteria. During the full-text screening, three additional studies were included through manual references checking, while ninety-six studies were excluded for reasons such as lacking numerical performance metrics ( $n = 16$ ), not using 3D LiDAR capture motion information ( $n = 40$ ), not involving human motion ( $n = 37$ ), not written in English ( $n = 2$ ), and being unavailable ( $n = 1$ ). Thirty-eight studies were included for data extraction (Fig. 1). The methodological quality scores of included studies ranged from 13 to 18, with an average score of 16.4 (Fig. 2).

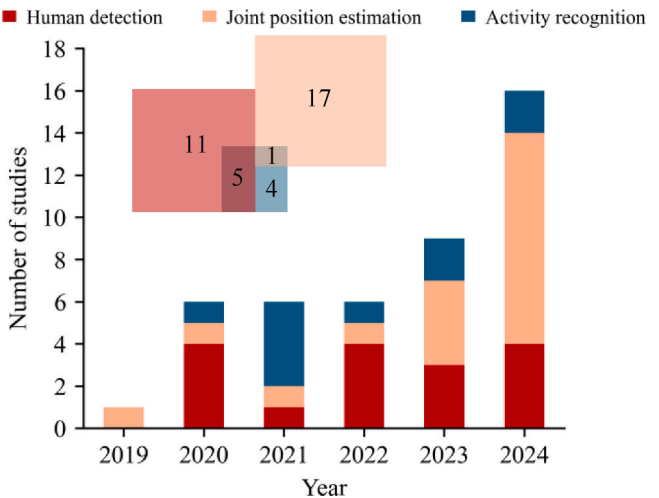
Thirty studies were of high quality, eight studies were of moderate quality and no study was of low quality. Studies scored highest on questions related to aims, datasets, experimental design, methodology, and evaluation metrics (Questions 1, 2, 4, 5, and 6), with all studies achieving full scores in these areas. However, validation, benchmarking,

and reproducibility were weaker, with average scores of 1.6 for validation (Question 7), 1.8 for comparisons with existing methods (Question 8), and 1.1 for code availability (Question 9). The number of relevant articles published increased on a yearly basis between 2019 (1 study) and 2024 (16 studies) (Fig. 3).

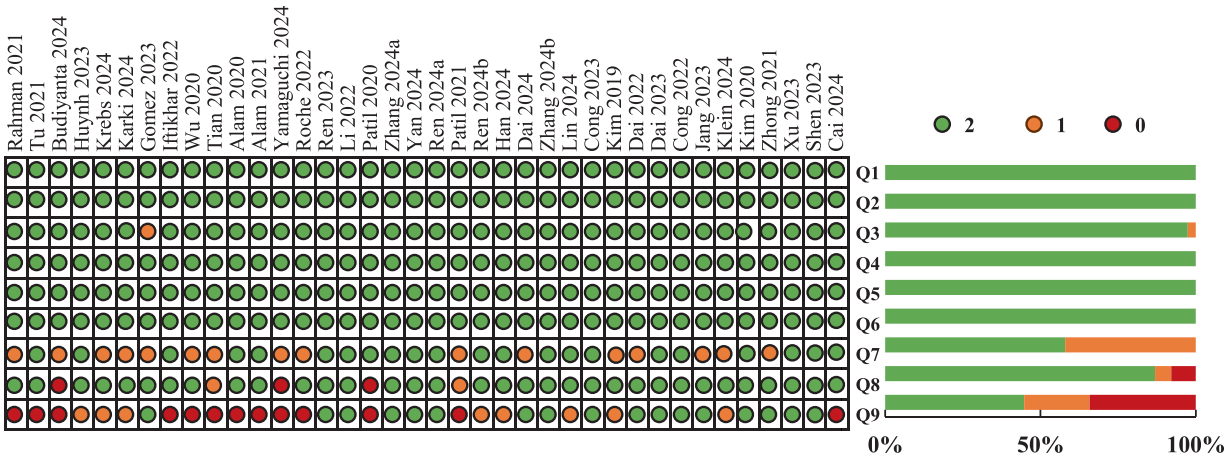
**3.2. LiDAR data preprocessing techniques**

Data preprocessing is a critical first step in LiDAR analysis to filter out non-human or irrelevant data and retain useful human-related point cloud information. The most common LiDAR data preprocessing techniques are (i) background subtraction, (ii) clustering, and (iii) point cloud resampling, which are performed sequentially.

Background subtraction involves removing static objects in the environment and isolating human motion data for further processing. A common approach is to first capture the static background point cloud in an environment without participants. Subsequent frames containing human motion are then processed by subtracting this pre-recorded background information, effectively eliminating stationary objects and retaining only the dynamic points corresponding to human movement (Karki et al., 2024; Kim et al., 2019; Tian et al., 2020; Ul Alam et al., 2021; Yamaguchi et al., 2024; Alam et al., 2020; Gomez et al., 2023; Li et al., 2022; Patil et al., 2020; Patil et al., 2021). This method is constrained by the need to pre-record background data in advance.



**Fig. 3.** Distribution of the included articles by year of publication.



**Fig. 2.** Results of the quality assessment based on nine questions for each reviewed study. Scores of 2, 1, and 0 are shown in green, orange, and red, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Following background subtraction, clustering is typically performed to segment the point cloud into distinct clusters, separating individuals or body parts from the background. To achieve this, established methods include the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) clustering algorithm (Lin et al., 2024; Ul Alam et al., 2021; Alam et al., 2020), Euclidean distance-based segmentation algorithm (Wu et al., 2020; Yamaguchi et al., 2024), mean shift algorithm (Karki et al., 2024). Following clustering, the human-related points are resampled in order to standardize the number of points per frame, ensuring consistent input dimensions across frames. The farthest point sampling algorithm is widely adopted to standardize the number of points per frame (Han et al., 2024; Ren et al., 2023, 2024a,b).

If LiDAR is fused with other devices, preprocessing typically requires multi-sensor coordinate calibration and time synchronization. When a LiDAR sensor operates in conjunction with IMUs or video cameras, coordinate calibration is necessary to align all sensor data within a unified reference frame. This can be achieved through geometric alignment using known inter-sensor spatial relationships (Cong et al., 2023; Krebs et al., 2024; Roche et al., 2022), or by using a known initial pose of a person (Dai et al., 2022). For accurate time synchronization, previous studies have employed calibration motions to align sensor data (Dai et al., 2022; Yan et al., 2023; Yan et al., 2024), as well as hardware-based precision time protocols (Lin et al., 2024).

### 3.3. Human detection

Human detection using LiDAR data is the process of identifying and localizing the presence of one or more people within the sensor's field of view. A total of 16 studies investigated human detection using LiDAR (Budiyaanta et al., 2024; Dai et al., 2022; Huynh and Alaghband, 2023; Iftikhar et al., 2022; Karki et al., 2024; Kim et al., 2020; Krebs et al., 2024; Roche et al., 2022; Tian et al., 2020; Ul Alam et al., 2021; Xu et al., 2023; Yamaguchi et al., 2024; Zhong et al., 2021; Alam et al., 2020; Cong et al., 2022; Gomez et al., 2023). Within these studies, the KITTI dataset was most widely used to test and validate human detection algorithms for activities performed in outdoor environments (Budiyaanta et al., 2024; Huynh and Alaghband, 2023; Iftikhar et al., 2022; Tian et al., 2020). Data fusion strategies were employed in six human detection studies to process multiple input data modalities (Fig. 4a). Specifically, five studies integrated LiDAR data with RGB images (Huynh and Alaghband, 2023; Iftikhar et al., 2022; Krebs et al., 2024; Roche et al., 2022; Tian et al., 2020), while one study combined LiDAR with IMU data for enhanced detection performance (Dai et al., 2022). To estimate human detection from LiDAR data, three studies utilized traditional machine learning techniques (Ul Alam et al., 2021; Yamaguchi et al., 2024; Alam et al., 2020), while eleven studies adopted deep learning methods (Budiyaanta et al., 2024; Cong et al., 2022; Dai et al., 2022; Gomez et al., 2023; Huynh and Alaghband, 2023; Iftikhar et al.,

2022; Karki et al., 2024; Kim et al., 2020; Roche et al., 2022; Tian et al., 2020; Zhong et al., 2021) (Fig. 4b). CNNs were the most commonly employed deep learning model for human detection from LiDAR data.

The most commonly used metrics to quantify human detection accuracy were precision (Gomez et al., 2023; Iftikhar et al., 2022; Karki et al., 2024; Roche et al., 2022), recall (Cong et al., 2022; Gomez et al., 2023; Iftikhar et al., 2022; Karki et al., 2024; Roche et al., 2022), and F1-score (Gomez et al., 2023; Karki et al., 2024; Roche et al., 2022) (Supplementary Material, Table S1). The best human detection performance was achieved using a CNN-based detection model YOLOv3, which reported an average precision of 97.7% and a recall score of 97.6% (Iftikhar et al., 2022). The highest F1-score of 0.95 was achieved using a geometric alignment approach that fused RGB images with LiDAR data (Roche et al., 2022).

Several approaches have been proposed for multi-person detection and tracking using LiDAR data (Budiyaanta et al., 2024; Huynh and Alaghband, 2023; Iftikhar et al., 2022; Karki et al., 2024; Krebs et al., 2024; Tian et al., 2020; Ul Alam et al., 2021; Yamaguchi et al., 2024; Alam et al., 2020; Cong et al., 2022; Gomez et al., 2023). Among them, one study addressed multi-person tracking by combining DeepSORT with Forest Path Cutting to delineate consistent motion trajectories for each person (Krebs et al., 2024). Another study adopted Global Nearest Neighbour data assignment, achieving precision of 96.4%, 97.5%, and 91.2% for three-person walking, complex poses, and occlusion scenarios, respectively (Gomez et al., 2023). To further address multi-person tracking, an Adaptive Order Hidden Markov Model combined with a Crossover Path Disambiguation Algorithm achieved a maximum Euclidean distance of 11.6 across individuals in different environments (Ul Alam et al., 2021). Another study utilized a Kalman filter-based approach to predict individual positions to maintain identity consistency across frames, reaching an average precision of 97.3% (Yamaguchi et al., 2024). In contrast, some studies performed only frame-by-frame detection in multi-person datasets without leveraging temporal continuity for identification or tracking. These studies relied on density-based clustering (Karki et al., 2024; Alam et al., 2020; Cong et al., 2022) integrating CNNs with image data (Iftikhar et al., 2022).

### 3.4. Human joint position estimation

Human joint position estimation from LiDAR data seeks to determine the 3D positions and orientations of anatomical joints from sensor data. A total of 18 studies performed human joint position estimation from LiDAR data (Cai et al., 2024; Cong et al., 2023; Dai et al., 2022, 2023, 2024; Han et al., 2024; Jang et al., 2023; Kim et al., 2019; Lin et al., 2024; Ren et al., 2023, 2024a,b; Zhang et al., 2024a,b; Klein et al., 2024; Li et al., 2022; Patil et al., 2020; Yan et al., 2024). The LidarHuman26M dataset was most commonly used for joint position estimation during activities performed in outdoor environments (Li et al., 2022). Over half

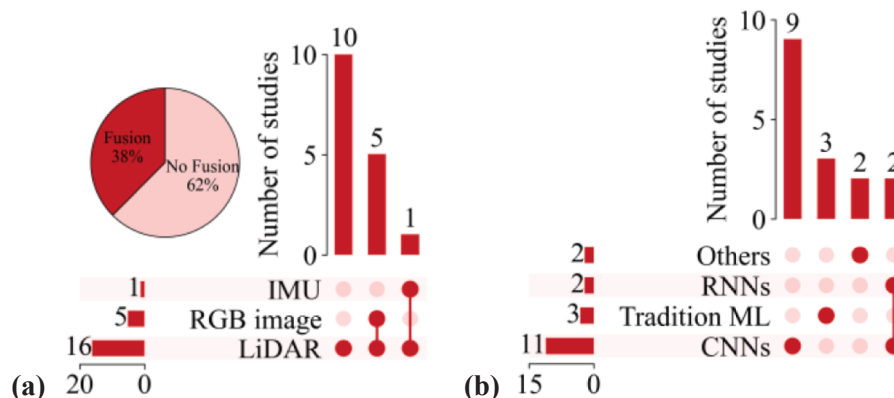


Fig. 4. Combinations and counts of human detection approaches across (a) input data types and (b) method categories.

of the reviewed studies performed multi-modal data fusion by combining LiDAR data with complementary sensing modalities such as IMUs or RGB images (Fig. 5a). Specifically, four studies fused LiDAR data with IMU data (Dai et al., 2024; Patil et al., 2020; Patil et al., 2021; Ren et al., 2023), five studies combined LiDAR with RGB images (Cai et al., 2024; Cong et al., 2023; Kim et al., 2019; Klein et al., 2024; Lin et al., 2024), while another study employed a fusion of LiDAR, RGB images and event-based data to enhance joint position estimation (Yan et al., 2024).

To estimate joint positions from LiDAR data, deep learning models were the most commonly used, adopted in 13 studies, with their use increasing substantially since 2022 (Cai et al., 2024; Cong et al., 2023; Han et al., 2024; Jang et al., 2023; Lin et al., 2024; Ren et al., 2023, 2024a,b; Zhang et al., 2024a,b; Klein et al., 2024; Li et al., 2022; Yan et al., 2024) (Fig. 5b). These deep learning models encompassed a variety of architectures, including RNNs, multi-layer perceptrons (MLPs), GNNs, and Transformers. Despite architectural differences, most deep neural network approaches use PointNet-family encoders for feature extraction, predominantly PointNet (Cong et al., 2023; Han et al., 2024; Ren et al., 2023, 2024a,b; Qi et al., 2017a) and PointNet++ (Cai et al., 2024; Jang et al., 2023; Lin et al., 2024; Qi et al., 2017b; Zhang et al., 2024a,b; Li et al., 2022; Yan et al., 2024). These encoders relied on symmetric aggregation functions to handle the unordered nature of point clouds. They serve as the backbone of current LiDAR-based human joint position estimation pipelines, supporting various temporal and spatial modelling strategies (Qi et al., 2017b; Qi et al., 2017a).

To capture motion dynamics using LiDAR data, a number of temporal modelling strategies have been adopted. RNN-based models, such as bidirectional gated recurrent units (biGRU), were commonly used to extract frame-to-frame temporal dependencies (Cai et al., 2024; Cong et al., 2023; Lin et al., 2024; Ren et al., 2023, 2024a; Zhang et al., 2024a; Li et al., 2022; Yan et al., 2024). To capture the spatial structure of the human body, GNNs-based models, including spatio-temporal graph convolutional networks (ST-GCN), were used (Cai et al., 2024; Jang et al., 2023; Li et al., 2022; Lin et al., 2024; Yan et al., 2024; Zhang et al., 2024a; Zhang et al., 2024b). More recent studies have employed transformer-based models to capture long-range spatial-temporal relationships by using attention mechanisms (Cong et al., 2023; Han et al., 2024; Jang et al., 2023; Ren et al., 2024a,b; Zhang et al., 2024b; Yan et al., 2024). Finally, MLPs were used as output heads to predicted joint motion using extracted features (Cai et al., 2024; Cong et al., 2023; Han et al., 2024; Jang et al., 2023; Lin et al., 2024; Ren et al., 2023, 2024a,b; Zhang et al., 2024a; Li et al., 2022).

For joint position estimation accuracy, mean per joint position error (MPJPE) (Cai et al., 2024; Cong et al., 2023; Dai et al., 2023; Han et al., 2024; Jang et al., 2023; Kim et al., 2019; Lin et al., 2024; Ren et al., 2023, 2024a; Zhang et al., 2024a,b; Li et al., 2022), percentage of correct keypoints (PCK) (Cong et al., 2023; Li et al., 2022; Yan et al., 2024;

Zhang et al., 2024a; Zhang et al., 2024b), and acceleration error (Cai et al., 2024; Ren et al., 2024b; Zhang et al., 2024a,b; Li et al., 2022; Yan et al., 2024) have been employed (Supplementary Material, Table S1). Across all studies included in this review, model performance using the MPJPE metric ranged from 30.0 mm to 119.3 mm (Fig. 6a). Among the studies that reported MPJPE, LiDARHuman26M and SLOPER4D were the most widely adopted benchmark datasets for joint position estimation using LiDAR data. Specifically, six models were evaluated on LiDARHuman26M datasets, including LiDARCap (baseline) (Li et al., 2022), LEIR (Yan et al., 2024), LiDAR-HMP (Han et al., 2024), LiDAR-CapV2 (Zhang et al., 2024a), Neighborhood-enhanced LidarCap (Zhang et al., 2024b), FineTemporalPose (Cai et al., 2024) and LIP (Ren et al., 2023). These models evaluated on the more challenging SLOPER4D dataset, which contains outdoor scenes, included the baseline (Dai et al., 2023), PEAR-Fuse (Lin et al., 2024), Neighborhood-enhanced LidarCap (Zhang et al., 2024b), LiDARCap (Dai et al., 2023; Li et al., 2022), LiveHPS (Ren et al., 2024a), and LiveHPS++ (Ren et al., 2024b).

The LIP model, a PointNet-based LiDAR-IMU fusion framework that integrates point cloud features with IMU signals from the wrist and ankle joints, achieved the best MPJPE performance when evaluated on the LiDARHuman26M benchmark dataset across different model comparisons (Ren et al., 2023) (Fig. 6b). It had the lowest MPJPE on simulated point clouds from the TotalCapture dataset (30.0 mm) (Fig. 6a). The LiveHPS++ model achieved the lowest MPJPE on the SLOPER4D dataset (77.0 mm), which was the second most commonly used benchmark (Ren et al., 2024b) (Fig. 6b). As a purely point-cloud-based model, LiveHPS++ incorporated a noise-insensitive velocity predictor to regress human velocity, further enhancing its performance on noisy input data.

Among included studies, the LiDARCapV2 model achieved the best PCK performance (96.5%), employing a hybrid architecture that combined PointNet++ with biGRU, MLPs, ST-GCN (Yu et al., 2018), and a skinned multi-person linear model for joint position estimation (Zhang et al., 2024a), (Loper et al., 2015). The LEIR model, which utilized a cross-attention fusion strategy, achieved the lowest mean acceleration error across human joints (23.9 mm/s<sup>2</sup>) (Yan et al., 2024). This approach also incorporated data fusion with RGB camera images and event-based sensor data to enhance temporal smoothness and motion consistency.

### 3.5. Human daily activity recognition

Human daily activity recognition seeks to identify and classify everyday human actions, including walking, sitting, drinking, and picking up objects. A total of 10 studies performed human daily activity recognition using LiDAR data (Lin et al., 2024; Rahman et al., 2021; Roche et al., 2022; Shen et al., 2023; Ul Alam et al., 2021; Wu et al., 2020; Xu et al., 2023; Yamaguchi et al., 2024; Alam et al., 2020; Tu

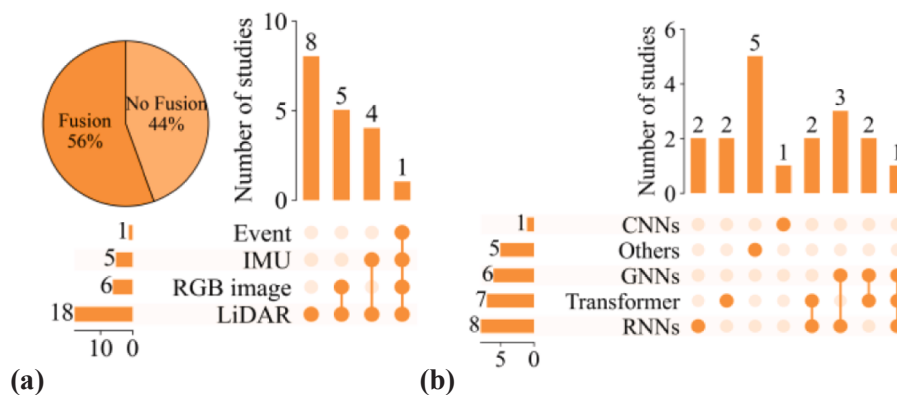
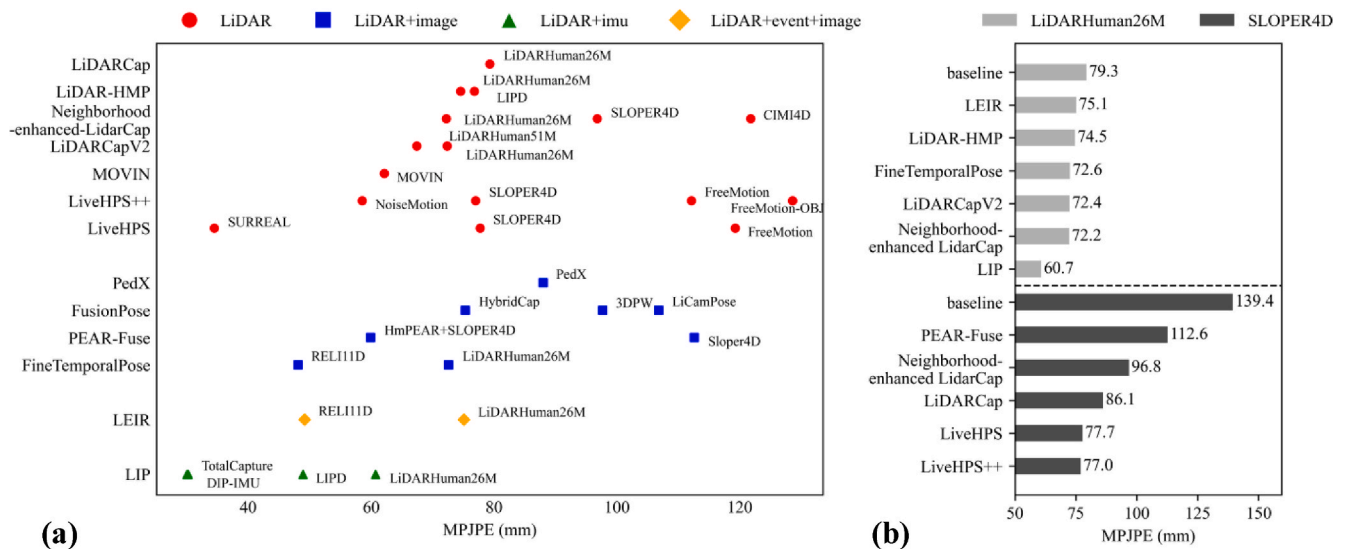


Fig. 5. Combinations and counts of human joint position estimation approaches across (a) input data types and (b) method categories.



**Fig. 6.** Performance comparison of state-of-the-art joint position estimation methods, showing (a) mean per joint position error (MPJPE) results of all models, and (b) MPJPE results on the LiDARHuman26M and SLOPER4D datasets.

et al., 2021). To achieve this, two studies combined LiDAR data with RGB images (Lin et al., 2024; Roche et al., 2022), while another integrated LiDAR with radar data to enhance activity classification performance (Rahman et al., 2021) (Fig. 7a). To perform activity recognition from LiDAR data, two studies utilized traditional machine learning techniques (Wu et al., 2020; Yamaguchi et al., 2024), while eight studies adopted deep learning methods (Lin et al., 2024; Rahman et al., 2021; Roche et al., 2022; Shen et al., 2023; Ul Alam et al., 2021; Xu et al., 2023; Alam et al., 2020; Tu et al., 2021) (Fig. 7b).

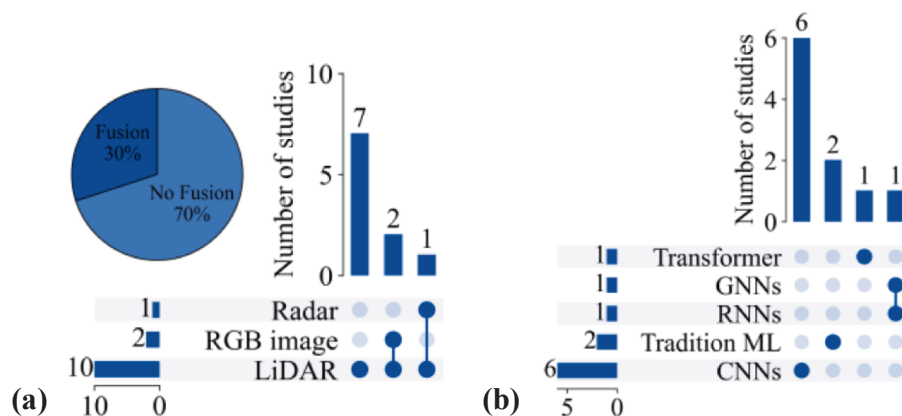
For human daily activity recognition, activity classification accuracy was the predominant metric employed to evaluate model performance (Lin et al., 2024; Rahman et al., 2021; Roche et al., 2022; Ul Alam et al., 2021; Yamaguchi et al., 2024; Alam et al., 2020; Tu et al., 2021) (Supplementary Material, Table S1). The highest human activity recognition accuracy was 98%, obtained using the CNNs-based Federated Framework with Dynamic Layer Sharing (Tu et al., 2021). Human keypoints have been extracted from point clouds and used as inputs for classifying 40 types of daily activities, including walking, standing, picking up objects, and drinking (Lin et al., 2024). In this study, a ST-GCN framework was employed to capture spatial dependencies among skeletal joints as well as temporal dynamics across consecutive frames, thereby enabling an explicit representation of inter-joint topological relationships.

### 3.6. Occlusion handling

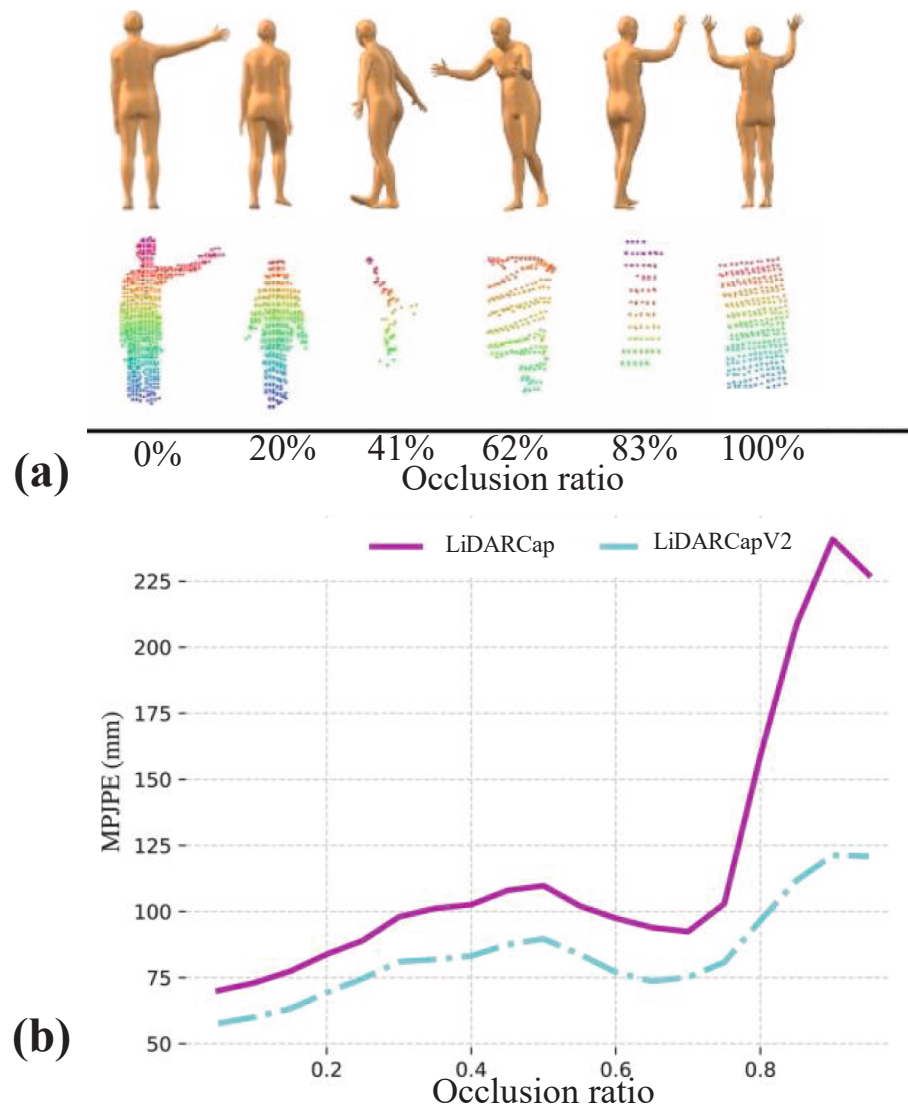
Occlusion in LiDAR-based human motion analysis results from blocking of the sensor’s straight-line of sight, which may result in incomplete or missing motion measurements (Fig. 8a and Fig. 8b) and can be categorized into self-occlusion and external occlusion. In experiments evaluating human detection under different levels of occlusion, detection performance decreased notably when more than half of the body is occluded, with average Recall dropping from 0.902 to 0.782 (Cong et al., 2022).

In addition to occlusion, human point clouds can also be affected by environmental noise from nearby objects, walls, and furniture. For example, when a person stands too close to an object and a point cloud from the two surfaces cannot be cleanly delineated. Noise mitigation in human point cloud data has been achieved by employing pre-segmented that excluded surrounding objects, walls, furniture, and other environmental noise. While this assumption simplifies processing, it reduces the robustness of these methods, particularly in the presence of background objects (Ren et al., 2024b).

To improve robustness to noise-contaminated data, a noise-insensitive approach LiveHPS++ utilized a cross-attention mechanism to associate each human joint with the most relevant point features, minimizing the impact of extraneous noise (Ren et al., 2024b). This study showed that blending synthetic noise-contaminated data with real



**Fig. 7.** Combinations and counts of human daily activity recognition approaches across (a) input data types and (b) method categories.



**Fig. 8.** Impact of occlusion on LiDAR data (Zhang et al., 2024a), illustrating (a) sample point clouds under different occlusion levels and (b) the corresponding joint position prediction errors.

samples improves model robustness. Increasing a synthetic data proportion from 0% to 100% improved robustness, lowering jitter from 68.8 to 30.9 and vertex error from 166.3 to 136.9. The LiDARCapV2 model, which utilized noise augmentation in its AgNoise-Segment module, achieved reduced sensitivity to noise in human point clouds (Zhang et al., 2024a). The effectiveness of this approach was evaluated at different noise levels (0.1, 0.2, 0.3, 0.4), yielding MPJPE values of 71.3, 72.0, 71.4, and 72.9 mm, respectively. These findings indicate that both occlusion and noise remain critical challenges for LiDAR-based human sensing, and that robustness can be significantly improved through noise-aware modeling and progressive synthetic data augmentation.

#### 4. Discussion

The purpose of this study was to review strategies for human motion analysis using 3D LiDAR, with a focus on methodologies for human detection, joint position estimation, and daily activity recognition. Despite its potential as a markerless and non-invasive sensing modality for capturing full-body movement in three dimensions, LiDAR-based motion analysis faces inherent challenges arising from the sparse, irregular, and unordered characteristics of point cloud data. Recent

advances in deep learning have supported the processing of such data from LiDAR, leading to significant improvements in estimates of human motion.

In LiDAR-based human detection studies, background subtraction and clustering-based segmentation methods have required pre-recorded background data without people in the field of view, which limits applicability in dynamic, complex multi-person environments encountered in the real world. Recent studies have explored multimodal fusion of LiDAR point clouds and RGB images, together with deep learning frameworks such as CNNs, to improve the robustness of human detection in diverse scenes (Huynh and Alaghband, 2023; Iftikhar et al., 2022; Krebs et al., 2024; Roche et al., 2022; Tian et al., 2020). In LiDAR-based joint position estimation studies, hybrid architectures combining the PointNet family of artificial neural networks, RNNs, and MLPs are commonly adopted to capture spatial and temporal features. For activity recognition studies, LiDAR-based approaches relied on inputting raw or preprocessed point clouds directly into classification networks. The sparsity of the point clouds obtained using LiDAR is affected by factors such as the sensing distance and the LiDAR model employed. Data processing methods demand considerable computational resources. Deep learning has increasingly been used to estimate human joint position from LiDAR data, providing accurate 3D joint motion data, and

enhancing activity classification models (Lin et al., 2024).

Marker-based optical motion capture and IMUs rely on wearable retro-reflective markers or sensors, while markerless video-based approaches are susceptible to ambient lighting conditions and may also raise privacy concerns due to photographic image recording. In contrast, LiDAR provides a markerless, non-invasive 3D full-body motion analysis solution robust to environmental lighting. Despite variability introduced by different datasets and model designs, the reported performance ranges of LiDAR-based approaches are broadly comparable to those of other motion capture technologies under similar applications (Table 2). For example, across LiDAR-, optoelectronic motion capture-, IMU-, and markerless video-based approaches, activity recognition accuracies exceeding 90% have been reported. Given the capacity of LiDAR to be used for three-dimensional motion analysis of the body with comparable activity recognition accuracy of up to 98% and joint center position errors below 30 mm, it may represent a promising and scalable direction for future motion analysis systems both in an out of laboratory settings. A number of LiDAR data processing techniques for human motion analysis have limitations that ought to be considered in future applications.

First, the sparsity of point clouds around distal joints such as the hands and feet leads to insufficient spatial sampling for reliable motion measurement. Owing to their small surface area, these distal joints often generate insufficient point cloud density, making precise localization of these anatomical structures difficult. As a result, reliable capture of distal joint motion remains a significant challenge, particularly under real-world conditions. A comparative analysis of recent studies reveals that multi-modal fusion, particularly the integration of 3D LiDAR with IMUs, can reduce the MPJPE from over 100 mm to approximately 30.0 mm (Ren et al., 2023; Zhang et al., 2024b).

Second, anatomically defined segment-specific coordinate systems are rarely implemented, which has limited the application of LiDAR-based approaches in standardized biomechanical motion analysis, including those that reference International Society of Biomechanics (ISB) recommended approaches (Wu and Cavanagh, 1995). Kinematics metrics such as joint angles and range of motion require anatomically defined segment-specific local coordinate systems. Although deep learning-based hybrid architectures achieve relatively accurate joint center position estimates from LiDAR point cloud data, most studies lack implementation of segment-specific coordinate systems. This inherently restricts the adoption of LiDAR technology in formal anatomical human motion analysis studies using ISB conventions. For instance, identifying a squat under fatigue conditions requires capture of subtle changes in joint angular velocities and compensatory movement patterns (Erman et al., 2023), including hip joint angle differences of approximately 6°. Current Li-DAR data processing models lack the sensitivity required to detect subtle functional declines or injury risks that are the primary

focus of formal human motion studies.

Third, processing 3D point cloud data for human motion analysis requires substantial computational resources. Despite achieving the 97.7% human detection precision and 98% activity recognition accuracy, raw LiDAR point clouds contain tens of thousands of points per frame, which is substantially higher than image- or IMU-based representations at the single-frame level. Moreover, over-reliance on raw point cloud data may have implications for data storage over time. For example, a 60 s recording captured at approximately 10 Hz with around 80,000 points per frame requires close to 1 Gigabyte of storage space. Data storage limitations ultimately impact the deployment of LiDAR-based systems in portable edge devices (e.g., resource-constrained embedded systems).

Fourth, measurement of human motion in multi-person scenarios remains challenging due to occlusion and intersection of person-specific motion trajectories, which can cause subject identity switching and poor joint localisation. While many LiDAR-based human motion analysis studies and public datasets are available, most data are collected under controlled conditions with limited multi-person interactions, which constrains benchmarking of robustness and generalization in more complex environments (Supplementary Material, Table S2) (Dai et al., 2023; Geiger et al., 2013; Fan et al., 2023; Li et al., 2022). A small number of recent studies have begun to address these challenges by incorporating attention mechanisms and noise augmentation strategies to improve robustness in crowded and noisy scenarios; however, systematic evaluation of these approaches in realistic settings and environments remains limited to date. This includes multi-person scenarios such as those commonly encountered in the clinic, free-living, and industrial environments.

These limitations highlight several directions for future research. Multimodal fusion between LiDAR and IMU sensors represents a promising approach to overcome limitations of geometry-based point cloud analysis, particularly in the presence of occlusion. Future studies ought to focus on integrating joint coordinate systems into LiDAR models to support standardisation of movement analysis and implementation of standard joint motion conventions (Wu and Cavanagh, 1995). Finally, future research ought to improve computational efficiency of point cloud processing, for example, through data compression or by replacing raw point cloud data with biomechanical parameters (e.g., joint angles, angular velocity, joint center positions) (Gao et al., 2025).

5. Conclusion

LiDAR technology provides a contactless, three-dimensional full-body human motion capture solution. Deep learning approaches, particularly the PointNet family of neural networks, are widely

**Table 2**  
Comparison of absolute error, mean per joint position error and activity classification accuracy for LiDAR and other established motion capture technologies.

|                                      | Joint center position detection |                                                           | Reference                           | Activity recognition |                                                    |
|--------------------------------------|---------------------------------|-----------------------------------------------------------|-------------------------------------|----------------------|----------------------------------------------------|
|                                      | Evaluation metric               | Errors                                                    |                                     | Evaluation metric    | Performance                                        |
| <b>Optoelectronic motion capture</b> | Absolute error                  | ~ 10 mm – 50 mm (Dunn et al., 2026; Miranda et al., 2013) | Bi-plane fluoroscopy                | Accuracy             | ~ 82.3% – 90.6% (Basak et al., 2022)               |
|                                      | MPJPE                           | ~ 26–107 mm (Niu et al., 2024)                            | Optoelectronic motion capture       | Accuracy             | ~ 89.5%–96.2% (Zhang et al., 2022)                 |
| <b>IMUs</b>                          | MPJPE                           | ~ 7–124 mm (Niu et al., 2024)                             | Optoelectronic motion capture       | Accuracy             | ~ 52.1% – 98.8% (Sharma et al., 2022)              |
| <b>Markerless Video</b>              | MPJPE                           | ~ 30–139 mm (Dai et al., 2023; Ren et al., 2023)          | IMU / Optoelectronic motion capture | Accuracy             | ~ 73.8% – 98% (Alam et al., 2020; Tu et al., 2021) |
| <b>LiDAR</b>                         | MPJPE                           |                                                           |                                     |                      |                                                    |

Absolute error (mm): Absolute difference between predicted and ground-truth joint center positions.  
MPJPE (mean per joint position error, mm): the average Euclidean distance between the predicted joint positions and the ground-truth joint center positions across all joints.  
Accuracy (%): the proportion of correctly classified instances among all instances.  
Activity recognition performance was evaluated using labelled trials.

employed for LiDAR-based motion analysis due to their symmetric aggregation functions, which ensure permutation invariance when learning from unordered and sparse point clouds. The findings of this review suggest that LiDAR-based methods can achieve joint centre position estimation and activity recognition performance comparable to IMU-based systems, optical motion capture, and video-based approaches. Although LiDAR shows promise as a viable alternative for comprehensive human motion analysis in the field of biomechanics, several limitations ought to be addressed for future application, including implementing anatomically defined joint coordinate systems for accurate joint kinematics, enhancing computational efficiency, and increasing robustness in multi-person environments.

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## CRediT authorship contribution statement

**Jiaqi Lai:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Mohammad Yavari:** Writing – review & editing, Formal analysis. **Peter Vee Sin Lee:** Project administration, Supervision, Writing – review & editing. **David C. Ackland:** Conceptualization, Supervision, Writing – review & editing, Project administration.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbiomech.2026.113292>.

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